POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Physics

Semester: Fall

Year : 2017
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Why compound pendulum is preferred than simple pendulum? Obtain time period of a bar pendulum and hence show that centre of oscillation and centre of suspension can be interchanged. 9

   b) A sinusoidal wave travels along a string. The time for a particular point to move from maximum displacement to zero is 0.17 s. What are the 6

      i. Period and frequency?

      ii. The wavelength is 1.40m; what is the wave speed?

2. a) What is interference and why are coherent sources necessary for interference? Explain necessary theory for Newton's rings method to determine the wavelength of monochromatic light. How can we get the centre of the rings bright? 9

   b) Deduce the missing orders in Fraunhoffer's double slit diffraction experiment in which slit width a=0.08mm and opaque width b=0.16mm. 6

3. a) Define quadrupole moment and derive the expression for potential at a point when point does not lie along a axis of quadrupole. 9

   b) A parallel plate capacitor has circular plates of 8 cm radius and 1 mm separation. What charge will appear on the plates if a p.d. of 100 volt is applied? 6

4. a) Compare Ampere's law with Biot – Savert law. Use Biot – Savert law to calculate the magnitude of magnetic field strength due to current carrying solenoid at any point. 9

   b) Show that ratio of electric field strength and magnetic field strength is equal to speed of light hence calculate maximum value of magnetic field and average intensity of light, if the maximum electric field at a distance of 10m from an isotropic light source is 2 V/m, 6

5. a) Using Schrodinger's wave equation obtain energy and wave-function of a particle confined in infinite potential well. 9

   b) What are the 6

      i. Mean time between collisions?

      ii. The mean free path for free electrons in copper? (Given $n=8.4\times10^{28}m^{-3}$, $\rho=1.7\times10^{-8}\Omega m$, $v_{avg}=1.6\times10^{6}$ m/s.)

6. a) Discuss the LC oscillation. Hence prove that the frequency of oscillation is $f=\frac{1}{2\pi\sqrt{LC}}$. Is this real or ideal explain? 9

   b) Discuss the factor reverberation for quality control of sound in public buildings. How it is different from echo. 6

7. Write short notes on: (**Any two**) 2×5

   a) Optical fibre and its uses

   b) Explain knee voltage and junction breakdown

   c) Time constant in LR circuit